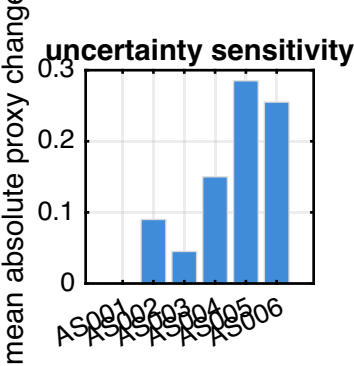
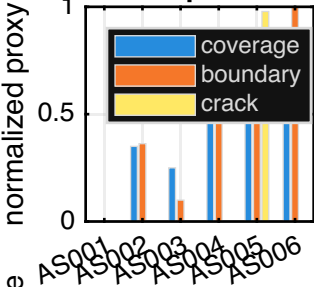
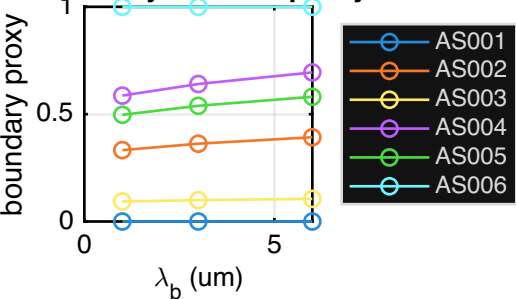


Phase 12B reduced geometry-driven mechanical forward model

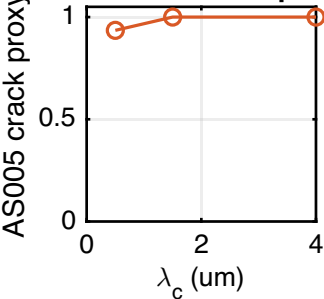
mechanical components by device



boundary-transfer proxy



crack-relaxation proxy



frozen model policy

- Common reduced components:**
- coverage_transfer
 - boundary_gradient
 - crack_relaxation
 - through_thickness_attenuation
- Raman-derived 2D field: blocked**
Transport relabeling: blocked

Phase 12B gates

